MAGI: state of validation

An internal reference for paper writing

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Status of this document

This is an internal working reference, not a manuscript and not submission-ready. It exists so that numbers entering the proceeding and the flash talk can be traced to the run that produced them.

Every numeric claim carries a provenance tag [Ex] resolving to Table 4. Tags point at files and runs on the local machine; they are *not* literature citations. This document contains no external references, because none have been verified for it.

Superseded and withdrawn results are retained and marked, not deleted. Work in progress is marked as such and carries no result.

1 Scope

MAGI is a conditional variational autoencoder that replaces Geant4 transport from a spacecraft or laboratory mass model down to a closed surface (the "CryoSphere") surrounding a cryogenic detector. Instead of tracking primaries through the full assembly, MAGI generates particle crossings directly on that surface; those crossings are then injected into the same mass model and transported the short remaining distance.

The validation question is therefore narrow and well posed: *does a MAGI crossing produce detector events at the same rate and with the same spectrum as a real recorded crossing, in the same geometry?*

Three independent sources have been tested: cosmic rays on the SRON X-IFU model (CR), cosmic rays on the DM1.2 laboratory cryostat, and ^{40}K radioactivity on the SRON model.

1.1 Model version under validation

All results below are **v0.8.2**, class `CVAE_MixEnergy_ContPhi_TaskAdaptive`: conditional rational-quadratic spline normalizing-flow continuum with CDF pre-warp, pinned 4 eV emission lines, focal-gated line/continuum supervision, and a learnable conditional coupling prior with zone conditioning. `latent_dim` = 8.

v0.8.3 (`energy_condition_geometry=True`, `latent_dim` = 12) was implemented and trained at three seeds and is **not adopted** (§8).

2 The normalization-free comparison

The central method, used for every headline number here:

$$R = \frac{N^{\mathrm{MAGI}}_{\mathrm{events}}/N^{\mathrm{MAGI}}_{\mathrm{crossings}}}{N^{\mathrm{real}}_{\mathrm{events}}/N^{\mathrm{real}}_{\mathrm{crossings}}} \tag{1}$$

Both sides are counted *per injected crossing*, through the identical mass model, binary, physics list and scoring code. R requires no exposure, no detector area, no flux normalization, and—critically—no crossing efficiency χ. It is insensitive to every quantity that has historically caused discordance between MAGI-side and Geant4-side bookkeeping.

The *efficiency factor* quoted in some tables is a different normalization, retained for continuity with earlier analysis:

$$\varepsilon \;=\; \frac{N_{\rm counts}^{\rm MAGI}/N_{\rm shot}^{\rm MAGI}}{N_{\rm counts}^{\rm ref}/N_{\rm shot}^{\rm ref}}, \qquad R = \varepsilon \times \chi, \tag{2}$$

where MAGI's shot particles are CryoSphere crossings and the reference's are thrown primaries. A perfect surrogate returns $\varepsilon = 1/\chi$, not 1.

3 Result 1 — Cosmic rays, DM1.2 laboratory cryostat

The cleanest measurement available, and the only one with a multi-seed ensemble at detector level.

Configuration. Reference: 2×10^8 muons thrown isotropically (`/gps/ang/type iso`, `mintheta 0`, `maxtheta 90`) through the DM1.2 mass model, yielding 999,358 CryoSphere crossings and 1,733 detector events [E1]. MAGI: 1,200,000 crossings generated from a 3-seed ensemble (seeds 42, 7, 13; two files per seed) and injected into the same model, yielding 2,019 detector events [E2]. Sphere: centre $(0, 0, 5)$ mm, $R = 95$ mm, verified from model metadata at generation time [E3].

Table 1: DM1.2: counts per injected crossing, MAGI vs full simulation.

Band	Reference	MAGI	R	
all deposits	1,733	2,019	0.970 ± 0.032	
100–300 keV (MIP)	1,325	1,589	$\mathbf{0.999 \pm 0.037}$	
300–1000 keV	310	323	0.868 ± 0.069	[E2]
30–100 keV	42	46	0.912 ± 0.195	
0.1–30 keV (X-IFU)	41	43	0.873 ± 0.191	
>1000 keV	13	17	1.089 ± 0.401	

Efficiency factor. $\chi_{\rm DM1.2} = 999{,}358/200{,}000{,}000 = 4.9968 \times 10^{-3}$, so a perfect surrogate returns $1/\chi = 200.13$ [E2].

Table 2: DM1.2 efficiency factor. The 0.1–30 keV row is the one directly comparable with the SRON CR value in §4.

Band	ε	ideal $1/\chi$	$R = \varepsilon\chi$	
0.1–30 keV (X-IFU)	174.8 ± 38.2	200.13	0.873	
1–20 keV	217.9 ± 56.8	200.13	1.089	[E2]
100–300 keV (MIP)	199.9 ± 7.4	200.13	0.999	
all deposits	194.2 ± 6.4	200.13	0.970	

What can and cannot be claimed. The MIP band is the spectral test: 1,325 reference events, ±2.7 %, and ε within 0.4 % of its ideal value. The X-IFU band is an *integrated rate only* — 41 reference events, ±15.6 % before MAGI contributes any error — and must not be presented as a spectral comparison. More MAGI generation cannot improve it; the reference is the cap.

DM1.2 has a **single** sensitive detector, so there is no across-detector spread to quote as a systematic; the ±38.2 is pure Poisson.

Structural checks (all pass). Crossing coverage 1,733/1,733 = 100.0 % [E2]; global species fractions correct to 0.3 % (γ 1.003, e^- 1.002, e^+ 1.039, μ^- 0.997) [E2]; impact-parameter distribution 0.912–1.010 over $b < 1$ to $b < 50$ mm [E2].

Known defect: a coherent 4 % deposit shift. A two-sample Kolmogorov–Smirnov test on all deposits gives $D = 0.0558$, $p = 0.0057$ [E2]. This is *not* a shape distortion but a uniform downward shift: deposit quantiles 25–95 % run 0.984, 0.960, 0.959, 0.940, 0.962, and the maximum CDF separation falls on the MIP peak at 186 keV. It localises to the MIP band ($D = 0.0594$, $p = 0.012$); the 300–1000 keV band and everything below 100 keV are indistinguishable ($p = 0.81$ and 0.99).

The cause is path length. Among detector-aimed crossings MAGI's tracks are too close to normal incidence: median $\sec\theta$ is 0.955 of the reference value, against a measured deposit-median ratio of 0.960 [E4]. Underlying it is a *direction-conditional species-mix error*: at $b < 10$ mm MAGI runs +8.8 % muons, −13.6 % e^-, −7.7 % γ, while global fractions are correct to 0.3 % [E4]. Per-species energies are correct (muons 0.99–1.03 at every cut).

Prerequisite finding: a geometry overlap that destroyed 39 % of crossings. At R = 105 mm the CryoSphere intersected `PlateCu1` (a 150 mm-radius, 5 mm-thick copper disc at z = 106.5, spanning $z \in [104, 109]$) by at least 1.29 mm — 13× the shell thickness — and was placed with overlap checking disabled. Geant4 navigation inside an overlap is undefined, and 39.4 % of detector events had *no recorded crossing*, uniformly 58–63 % across five independent runs, 93.5 % of them primary-muon deposits [E5]. Moving the sphere to R = 95 mm restores 100.0 % coverage. The SRON sphere was checked and is clean.

Separately, the DM1.2 muon macro had used `/gps/ang/type focused` ten times, aiming every muon at the origin (median cos 0.994 to the origin, 56.8 % inside cos > 0.99); absolute rates from that configuration were never physical [E5]. All DM1.2 results here are from the `iso` rerun.

4 Result 2 — Cosmic rays, SRON X-IFU model

Normalization-free. $R = \mathbf{0.848 \pm 0.125}$ [E6], from 51 counts in 1,600,000 injected real crossings against 500 counts in 18,500,000 injected MAGI crossings, same geometry and binary. (The uncertainty combines both arms; an earlier statement of ±0.14 used the reference arm alone.)

Per-detector flux ratio against the RUN CRYOAC reference, 1–20 keV: 0.76, 0.90, 0.85, 0.57, 0.81, 0.87 over detectors 1–6, mean **0.79** [E7]. Spectral shape is **indistinguishable** from the reference: KS $D = 0.048$, $p = 0.20$ [E7]. For contrast, v0.6.2 gave $p = 0.0014$.

Efficiency factor. Counts in 0.1–30 keV [E7]:

Table 3: SRON CR efficiency factor per detector, v0.8.2. Ideal value $1/\chi_{\rm CR} = 164.4$.

Detector	Reference counts	ε (v0.8.2)
1 — TES array	28,402	124.8 ± 5.6
2 — Back CryoAC	11,187	142.6 ± 9.6
3 — Lateral	2,922	121.3 ± 17.3
4 — Lateral	3,036	98.1 ± 15.2
5 — Lateral	2,829	122.8 ± 17.7
6 — Lateral	3,039	135.3 ± 17.9
mean		**124.1**
std across detectors		15.2

Normalized, $R = \varepsilon\chi = \mathbf{0.76}$, consistent with the 0.79 of the flux ratio and the 0.848 of the normalization-free arm, by three independent code paths.

The SRON–DM1.2 contrast. This is the most important open physics question in the validation:

	ε (0.1–30 keV)	ideal $1/\chi$	R
SRON CR, 6 detectors	124.1 ± 15.2	164.4	**0.76**
DM1.2, single detector	174.8 ± 38.2	200.13	0.873 ± 0.191
DM1.2, MIP band	199.9 ± 7.4	200.13	0.999 ± 0.037

DM1.2 sits at its ideal value where statistics are good; SRON shows a ∼24 % deficit that DM1.2 does not reproduce. The DM1.2 overlap bug was excluded as the cause — the SRON sphere does not overlap.

Identified contribution to the SRON deficit. An **inconsistent ingoing-particle cut** [E8]. `EventAction.cc` skips outgoing crossings at registration, but the SRON training sets predate that code: the CR training set contains 10.37 % outgoing crossings and the ^{40}K set 22.53 %, while the DM1.2 set contains 0.00 %. MAGI applies no ingoing/outgoing selection either, and generates 13.59 % outgoing against the training set's 10.37 % — accounting for 3.6 % of the rate deficit. This also means **SRON and DM1.2 models are on different conventions.**

5 Result 3 — ^{40}K, SRON BuildingModel

R = **1.118 ± 0.373, KS** D = 0.2384**,** p = 0.60 [E9]: 2×10^8 injected on both sides into `SRON_CCNCoolerwithXFDM_noShields_BuildingModel.gdml`, giving 19 MAGI counts (9.50×10^{-8} per crossing) against 17 from real recorded crossings (8.50×10^{-8}). Matched N by construction. **^{40}K passes at detector level.**

The real-crossing arm recycles 1,873,609 recorded crossings ∼107×, which is sound because Geant4 re-randomizes the interaction on each replay; continuity was verified from the crossing counts (∼40.15 M re-registered per job, 0.80 per injected event, uniform across four jobs) rather than assumed [E9].

The 22× gap is Geant4 bookkeeping, not MAGI. Against the 44-days reference the same run reads 0.049 ± 0.012. The reference and the MAGI training set *do not share a ^{40}K source*: the reference samples a $2000 \times 2000 \times 400$ mm volume centred at $z = -1600$ mm, the training set $750 \times 750 \times 50$ mm at $z = -1250$ mm — a **57**× smaller sampling volume [E10]. The crossing yield depends on source geometry and not on a scalar, so the two cannot be chained, and the existing training set cannot be renormalized into the reference. MAGI sits with the population it was trained on.

Bookkeeping correction. `NgenPotassiumForMAGI_v082` was declared 580,000,000; measured from the run's own output it is 538,205,667 [E11]. The output file concatenates two jobs, with exactly one backward `EventId` jump at row 398,400; estimating each job separately by $N = \max \cdot (n+1)/n$ gives 438,145,967 and 100,059,700. The declared value is 7.8 % high and outside the 95 % region of both jobs' order statistics.

The same estimator was run as a control on two other declared counts and *confirms* them (^{40}K v0.7.2 ratio 1.0001; CR v0.8.2 ratio 0.9998), so it is not systematically low. Every v0.8.2 ^{40}K *Flower* flux and efficiency rises by 1.0777; the Detector-1 efficiency goes 3472 → 3741. The detector-level result above is unaffected — it never uses the constant.

6 Cost — measured on DM1.2

The cost claim must not be an absolute wall-clock number, which depends on the machine. It must be a *ratio* with both sides measured on the same hardware, and it must be decomposed — because the naive figure is wrong by two orders of magnitude. DM1.2 is the only case where both sides ran here; the SRON equivalent **cannot** be computed, since that reference ran on a cluster [E15].

Step	wall	jobs	core-h	output
full simulation, 2×10^8 muons	12.71 h	8	**101.67**	999,358 crossings, 1,733 events
MAGI generation, 1.2×10^6	31 s	6	0.05	—
MAGI training, 3 seeds	486/620/743 s	1	0.51	—
MAGI injection, 1.2×10^6	9.74 h	6	**58.46**	2,019 events

The efficiency factor is not the speed-up. This is the trap, and it belongs in the paper explicitly because a referee will look for it.

Detector events per thrown muon	8.665×10^{-6}
Detector events per injected crossing	1.683×10^{-3}
Statistical amplification	**194.2**× (ideal $1/\chi = 200.13$)
core-h per thrown muon	5.084×10^{-7}
core-h per injected crossing	4.872×10^{-5}
Transport cost ratio	**95.8**×
Net throughput gain	**194.2/95.8 = 2.03**×

Cross-checked by equivalent statistics: 1.2×10^6 crossings carry 2.402×10^8 equivalent muons, so MAGI delivers 4.11×10^6 equiv-muons/core-h against the full simulation's 1.97×10^6 — 2.09×, consistent. A crossing costs ∼96× more to transport than a thrown muon *because it starts on the sphere, inside the assembly, in material and adjacent to the detector*, whereas most of the 2×10^8 thrown muons miss the apparatus and are killed almost immediately. MAGI removes the cheapest transport in the problem.

General form, and why DM1.2 is the pessimistic case.

$$\text{speed-up} = \frac{1}{\chi} \times \frac{c_{\text{primary}}}{c_{\text{crossing}}} \tag{3}$$

One full run decomposes into outer transport (primary → sphere, **eliminated** by MAGI) 53.0 core-h, 52 %; and inner transport (sphere → detector, **still paid**) 48.7 core-h, 48 %. The gain is set by how much material sits *outside* the CryoSphere relative to inside. DM1.2 is close to the worst case: a small laboratory cryostat with an $R = 95$ mm sphere enclosing most of the assembly. For a spacecraft primaries traverse metres of structure while the sphere-to-detector distance stays at centimetres, so the gain should be far larger — but **that has not been measured and must not be asserted.**

Reuse across design variants — where the speed-up lives. The decisive property is that the CryoSphere crossing population is determined *entirely by the geometry outside the sphere*. Design iteration changes the inside — detector layout, filters, readout electronics, shielding around the absorber — and leaves the outer structure untouched, so the training set stays valid across the whole campaign and the full simulation is paid **once**, not once per variant. For N inner-design variants at reference-equivalent statistics: full simulation $N \times 101.7$ core-h; MAGI $102.2 + N \times 48.7$ core-h.

N variants	full sim	MAGI	speed-up
1	102	151	0.67×
2	203	200	1.02×
5	508	346	1.47×
10	1,017	589	1.73×
20	2,033	1,076	1.89×
∞			**2.09×**

Break-even at $N \approx 1.93$ — the second design variant. This is the same arithmetic as "1.93× more statistics of the same geometry", but it means something entirely different: nobody wants to re-run an identical geometry for twice the statistics, whereas testing a second inner configuration is what a design campaign does on day one. For a flight model, where the outer share exceeds DM1.2's 52 %, the break-even approaches $N = 1$ — MAGI pays for itself on the *first* redesign.

The claim that does not depend on compute at all. MAGI converts a **statistically exhausted** sample into an unlimited one. The ^{40}K training set holds 1,873,609 real crossings and more cannot be had without re-running an enormous campaign; MAGI generated 538,205,667 from it, a **287× amplification**, validated at the detector at 1.118 ± 0.373. No amount of compute buys that from the original sample.

7 Correction: the "seed systematic" is withdrawn

Withdrawn claim

Earlier versions of the internal documents reported that MAGI's detector-relevant energy ratio spans 0.455–1.222 across training seeds, and recommended quoting a ±40 % single-seed / ±15 % ensembled systematic. **That spread is an artefact of the metric and is withdrawn.** It was never measured on a physical quantity.

The metric `energy_vs_impact_parameter` compares median generated to median real energy among detector-aimed crossings. On the CR source that statistic estimates the *species mix*, not the energy conditional.

At $b < 10$ mm the real population is [E12]: μ^- 48.008 % at median 828.373 MeV; γ 49.789 % at median 0.511 MeV; e^- 2.092 % at median 3.170 MeV; aggregate median 13.288 MeV. The two dominant populations are three orders of magnitude apart, split almost exactly 50/50, and *disjoint* (γ 99th percentile 30.488 MeV, μ^- 1st percentile 24.072 MeV). The 50th percentile of the combined sample therefore falls in the empty gap between them, and the aggregate median is close to a step function of the muon fraction.

Demonstration using the real data only. Reweighting the species mix of the real $b < 10$ mm crossings, with every energy value left untouched, so that the energy error is zero by construction [E12]:

μ^- reweight	μ^- fraction	aggregate median	ratio to truth
0.955	46.834 %	9.287 MeV	**0.699**
0.980	47.686 %	11.754 MeV	0.885
1.000	47.953 %	13.172 MeV	0.991
1.012	48.325 %	14.982 MeV	**1.128**
1.050	49.268 %	21.316 MeV	1.604

A $\pm 5\,\%$ mix shift swings the metric over 0.70–1.60. The measured per-seed muon reweights at $b < 10$ mm are 1.012/1.012/0.955 — precisely this range, and precisely the seeds that produced the reported spread.

The energy conditional, resolved per species [E12]:

Species	seed 42	seed 7	seed 13
γ	1.000–1.001	1.000–1.001	1.000–1.039
μ^-	0.956–1.003	1.043–1.064	0.947–0.976
e^-	0.680–1.303	0.805–1.005	0.555–0.615

(ranges over the cuts all-ingoing, $b < 50$, $b < 20$, $b < 10$ mm)

Gammas are reproduced *exactly* at every cut and every seed; muons within $\pm 6\,\%$. e^- scatters but is 1.9–2.1 % of the population with 639 real events at $b < 10$ mm — statistics-limited, not a demonstrated defect.

What is real. A $\pm 5\,\%$ seed variation in the species mix *conditional on direction* (μ^- reweight 1.012/1.012/0.955 at $b < 10$ mm, against global fractions correct to 0.3 %) [E12]. This is genuinely a joint-distribution failure that marginal validation cannot see — an order of magnitude smaller than claimed, and in a different channel. The same signature appears independently on DM1.2 (§3).

Consequence for §4. The §4b diagnostic in the validation note — "the model has invented an anti-correlation between energy and aiming" — rests on the same aggregate statistic and is **not currently supported**. Its marginals table, aimed-fraction column and counts are unaffected. The detector-level $R = 0.848 \pm 0.125$ never used the metric and stands.

8 v0.8.3: implemented, tested, not adopted

`energy_condition_geometry=True` plus `latent_dim` 8 → 12, trained at three seeds [E13]. It did not improve the target metric and **regressed the two lowest energy decades consistently across all three seeds**: 0.001–0.01 MeV from 0.880/0.921/0.995 to 0.677/0.715/0.738; 0.01–0.1 MeV from 1.686/1.296/1.767 to 2.216/1.714/1.979. That regression is measured on *marginals* and therefore survives the correction in §7; it remains a sufficient reason not to adopt v0.8.3.

Since §7 shows the energy conditional was not the broken term, `energy_condition_geometry` was aimed at a quantity that is not at fault, which explains why it did not help. The flag remains in `core/model.py` defaulting to `False`, so v0.8.2 is bit-identical and the experiment is resumable.

Revised priority for the next version.

1. Make `energy_vs_impact_parameter` species-resolved by default and refuse an aggregate on multi-species input.
2. Target the aimed species mix: condition the *type* draw on sampled geometry, rather than sampling from a global `type_probs`.
3. Measure the detector-level seed spread directly (in progress, §9).
4. More generated events at small b to settle whether the e^- scatter is real.

9 Work in progress — no result yet

Running as of 14 August 2026, 08:00 CEST

Three-seed detector-level CR comparison on SRON. Six jobs, two per checkpoint (seeds 42, 7, 13), 1,200,000 crossings each — verified exactly 1,200,000 records per file with no neutrinos — injected into `SRON_CCNwithXFDM_NoShield_FlowerCryoAC_fixed.gdml`, the same geometry as the arm-A reference [E14].

This is the measurement that has never been taken: every seed claim to date, withdrawn or not, is crossing-level. Expected sensitivity is ∼62 Detector-1 counts per seed (±13 %) and ∼112 over all six detectors (±9 %) — enough to exclude a factor-2 spread decisively, marginal at ±20 %, and **insufficient** to resolve the ±5 % aimed-mix effect. ETA ∼17 h.

No number from this run may be quoted until it completes.

10 What may and may not be quoted

Quotable

- CR, SRON, normalization-free: $R = 0.848 \pm 0.125$.
- CR, SRON, spectral shape: KS $p = 0.20$, indistinguishable from RUN CRYOAC.
- CR, SRON, efficiency: 124.1 ± 15.2 over 6 detectors ($R = 0.76$).
- DM1.2, 3-seed ensemble: $R = 0.970 \pm 0.032$ overall, 0.999 ± 0.037 in the MIP band, 100 % crossing coverage.
- DM1.2 efficiency: 174.8 ± 38.2 (X-IFU band), 199.9 ± 7.4 (MIP), against ideal 200.13.
- ^{40}K, BuildingModel: $R = 1.118 \pm 0.373$, KS $p = 0.60$.
- Species fractions reproduced to 0.3 %; impact-parameter distribution to within 2–9 %.
- Cost, DM1.2: **194**× statistical amplification per transported particle (ideal 200.13) and **2.03**× net throughput, both measured on the same machine; 52 %/48 % outer/inner cost split; break-even at the 2nd design variant.
- ^{40}K: **287**× amplification of an exhausted sample.

Not quotable

- **Any seed systematic**, in either direction. Not established.
- Any aggregate median-E-vs-b number (§7).
- The ^{40}K 44-days comparison (0.049) as a MAGI result — the source geometries differ by 57× (§4 above).
- DM1.2 X-IFU band as a *spectral* comparison — 41 reference events; it is an integrated rate.
- Impact-parameter bins below $b \approx 5$ mm at present sample sizes.
- Absolute rates from any DM1.2 `focused`-source run.
- **A** ∼200× **speed-up.** 200.13 is the statistical amplification, not the compute saving; the measured throughput gain is 2.03× (§6).
- Any numeric cost extrapolation to Athena/X-IFU — the outer/inner core-hour split has not been measured on a flight mass model.
- Absolute wall-clock timings as a cross-machine claim.

11 Open items

1. **The SRON 24 % deficit is not fully explained.** 3.6 % is attributed to the ingoing-cut mismatch (§4); the remainder is uncharacterized now that the energy×geometry diagnosis is withdrawn.
2. **Ingoing-only filter** on both SRON training sets, with χ recomputed, then retrain — makes SRON consistent with DM1.2.
3. **Detector-level seed spread** on CR (§9).
4. e^- **behaviour at small** b — needs generation, not seeds.
5. **Outer/inner core-hour split on a flight mass model.** DM1.2 is measured (§6); the Athena/X-IFU case — where the argument is strongest — is not, and is the one number that would let the cost claim be stated for a real mission.
6. `GeneratedParticlesRecycled` did not fire during ∼107× recycling — the warning path in `PrimaryGeneratorAction.cc` is unreachable. Silent recycling is exactly what it exists to catch.
7. **DM1.2 fluorescence lines** — the training set supports continuum and geometry but is ∼9× short for lines.

12 Provenance

Table 4: Evidence tags. All paths are local to the analysis machine.

Tag	Artifact
E1	`S1GDML_DM1.2/allOutputGDML_DM1_2_iso.dat`, `allDSCryoSphere_DM1_2_iso.dat` — the 2×10^8-muon iso reference run.
E2	`S1GDML_DM1.2/Analysis/MAGI_DM1_2_iso_comparison.ipynb`, executed 13/08/2026 22:57; jobs in `S1GDML_DM1.2/magi_iso_run/job0..5`. Figures: `MAGI_DM1_2_iso_{spectrum,crossings,aiming}.png`.
E3	`magi_iso_run/gen_s0..5.log` — the “CryoSphere for reconstruction” line emitted by `MAGI/scripts/generate_geant_source.py`.
E4	Incidence-angle and aimed-species analysis over `/Volumes/X10Pro/tmp/magi_dm12iso_s0..5.gntbin` against `allDSCryoSphere_DM1_2_iso.dat`, 13/08/2026.
E5	`S1GDML_DM1.2/CRYOSPHERE_OVERLAP_FIX.md`; `src/GDMLDetectorConstruction.cc` (R 105→95, overlap check on); `gpsMu_iso.mac`.
E6	`S1GDMLSRON_CryoSphereEmission/ab_run/` arm A vs the v0.8.2 1.85×10^7 run.
E7	`S1GDMLSRON_CryoSphereEmission/Analysis/DataAnalysis_pkg.ipynb`, sections *RUN CRYOAC vs MAGI* and *Efficiency factor per detector*; `docs/MAGI_validation_note.md` §2–§3.
E8	`docs/MAGI_validation_note.md` §5; `src/EventAction.cc` ingoing test applied to the training files.
E9	`k40_run/run_now.sh`, `k40_run/run_real_bm.sh`, `k40_run/compare_real_vs_magi.txt`.
E10	`Old-Simulations/Sim/S1GDMLSRON/gpsPotassium40.mac` vs `S1GDMLSRONBuilding/gpsPotassium40.mac`.
E11	`athena-build/outputGDML_K40_v082.dat` EventId analysis; `DataAnalysis_pkg.ipynb` cell 4; `MAGI_validation_note.md` §7.6.
E12	`MAGI/tools/species_resolved_evsb.py`, `MAGI/logs/species_evsb.log`, 13/08/2026; real reference `MAGI/TrainingData/alloutputDSCryoSphereCR.dat` (3,285,347 crossings, neutrino-free).

Tag	Artifact
E13	`MAGI/tools/train_v083_cr_seed.py`, `MAGI/tools/compare_v082_v083.py`, `MAGI/docs/v0.8.3_seed_variance.md`.
E14	`S1GDMLSRON_CryoSphereEmission/cr_seed_run/run_cr_seeds.sh`, launched 14/08/2026 07:54. **In progress.**
E15	Timings from run-directory timestamps and log headers: `S1GDML_DM1.2/iso_run/iso_run.log` (12/08 12:36:49 → 13/08 01:19:21, 8 jobs); `magi_iso_run/magi_iso.log` (13/08 13:12:42 → 22:57:20, 6 jobs); `MAGI/tools/dm12_iso_seed{42,7,13}.log` (486/620/743 s). Both Geant4 runs saturated the same 8-core machine (8 vs 6 jobs), so the per-core normalization carries a ±10–20 % systematic.

Companion documents.

- `S1GDMLSRON_CryoSphereEmission/docs/MAGI_validation_note.md` — the authoritative running record, including superseded sections.
- `MAGI/docs/v0.8.3_seed_variance.md` — the withdrawn seed systematic and the revised plan.
- `S1GDML_DM1.2/CRYOSPHERE_OVERLAP_FIX.md` — the overlap investigation.